Elias Pedroza

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Education

Northwestern University **Expected Dec. 2027**
Master of Science in Robotics *Evanston, IL*
- Coursework: Robotic Manipulation · Embedded Systems in Robotics (ROS 2) · Machine Learning · Machine Dynamics

Santa Clara University **Jun. 2026**
Bachelor of Science in Mechanical Engineering, cum laude *Santa Clara, CA*

Experience

1X Technologies **Mar. 2026 – Sep. 2026**
Robot Service Engineering Technician *SF Bay Area, CA*
- Diagnosed mechanical, electrical, & software failures via inspection, terminal commands, & digital twins
- Performed precision repairs on humanoid robots, interpreting CAD & electrical schematics to verify calibration & wiring
- Documented root causes & executed test sequences to verify system performance, managing full repair life cycles in Jira

Design for Assistive Robotic Technologies (DART) Lab **Jun. 2025 – Jun. 2026**
Research Intern · Co-author, IEEE-RAS Humanoids 2026 paper *Santa Clara, CA*
- Designed laminar-jamming prosthetic fingers that stiffen 2.3× under vacuum, with flexures cut 55% thinner
- Built 3-part silicone casting molds that fixed early vacuum leaks, plus tip-load rigs to compare flexure layer materials
- Developed a whiffletree that shares one harness cable across both fingers, letting them wrap around uneven objects

NASA L'SPACE Mission Concept Academy **Jan. 2025 – May 2025**
Thermal Engineer *Remote*
- Designed the thermal control system for a Venus aerobot, holding the payload near 300 K in a 225.8–344.7 K atmosphere
- Developed heat flow maps to model the payload steady-state behavior under solar, albedo, convection, & radiation loads
- Presented & co-authored a 191-page design review, coordinating subsystems with a 12-member multidisciplinary team

Projects

Brick-Catching Robot & Waypoint Tracking | *Embedded Systems in Robotics (Northwestern)*
- Built a ROS 2 brick-catching robot sim (URDF, tf2) that decides from fall time if a brick is reachable, then catches it
- Split physics, odometry-based control, & decision-making into separate nodes, validated by 6 automated tests
- Developed a 100 Hz waypoint controller with custom ROS 2 services, driving a 33 m loop within 1% of the ideal path
- Added a node injecting Gaussian noise into velocity commands & published per-lap distance, error, & time metrics

Integrated Reef Information System (IRIS) | *SCU Robotic Systems Lab (RSL) (Senior Design)*
- Developed an autonomous buoy–winch system that surfaces to relay reef sensor data, eliminating diver-based retrieval
- Designed a self-righting marine buoy (+31.5 mm metacentric height) with custom tether mounts & modular ballast
- Collaborated on a servo-driven winch with gear reduction & ratchet-pawl locking for tether-buoy depth control

GOAL! (Penalty Shootout Arcade Game) | *Advanced Mechatronics I (Graduate)*
- Designed & fabricated a 2-player electromechanical game featuring a 3D-printed rack & pinion kicker & servo goalie
- Programmed an Arduino Mega in C++ for game logic & real-time audio/visual feedback from joystick & sensor inputs
- Wired the optical coin detector, servos, joysticks, & displays, running a 6-hour public demo without interruption

Mobile Manipulator Maze Challenge | *Robotics Final Project*
- Programmed a LIMO Cobot mobile manipulator in ROS & Python, tuning MoveIt arm poses for pick-and-place
- Navigated to 3 of 4 QR codes, successfully picked up & delivered a sock to its target, earning joint-highest course score

Technical Skills

Software: Python · C · C++ · ROS 2 / ROS · MoveIt · MATLAB · Linux · Git

CAD & Simulation: SOLIDWORKS (Certified Associate) · Siemens NX · Ansys (FEA/CFD) · GD&T

Hardware: Arduino · 3D Printing (FDM, SLS) · Laser Cutting · Circuit Design · Soldering · Silicone Molding · Machining

Leadership & Activities

- Pi Tau Sigma Mechanical Engineering Honors Society
- Course Grader, Mechatronics: Haptics (Graduate)
- SHPE Community Outreach Director
- Summer Engineering Seminar Mentor